

Unit 6

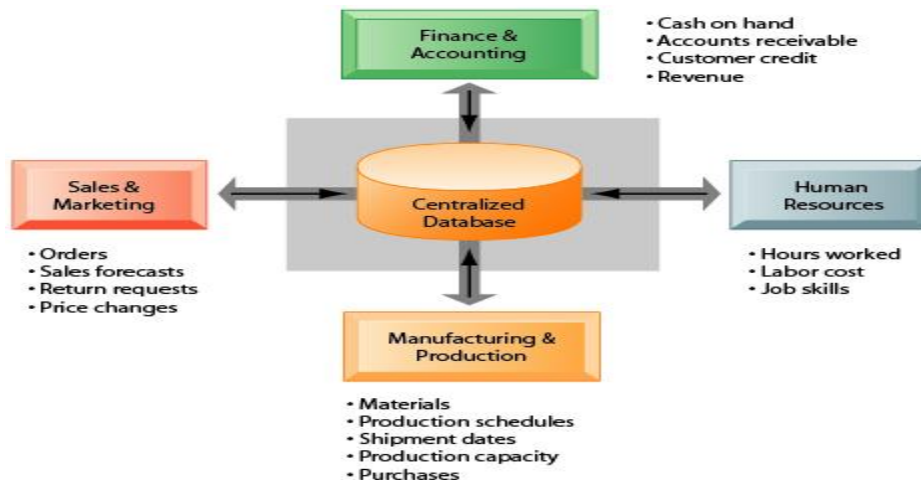
Key System Applications for Digital Age

Enterprise Systems

Enterprise systems or enterprise resource planning (ERP) systems, integrate the key internal business processes of a firm into a single software system so that information flows seamlessly throughout the organization, improving coordination, efficiency and Decision making.

Enterprise software is based on a suite of integrated software modules and a common central database. The database collects data from and feeds the data into numerous applications that supports nearly all of an organization's internal business activities. When new information is entered by one process, the information is made

available immediately to other business processes.



HOW ENTERPRISE SYSTEM WORK

Enterprise systems feature a set of integrated software modules and a central database that enables data to be shared by many different business processes and functional areas throughout the enterprise.

Enterprise software is built around thousands of predefined business processes that reflect best practices. **Best practices** are the most successful solutions or problem-solving methods in an industry for consistently and effectively achieving a business objective.

Organizations implementing commercial enterprise software first select the business processes they wish to use from the software and map their own processes to these, using the software's configuration tables. Although businesses may choose to rewrite

portions of the software to match their existing processes, this can degrade system performance and fail to reap the benefits of this software.

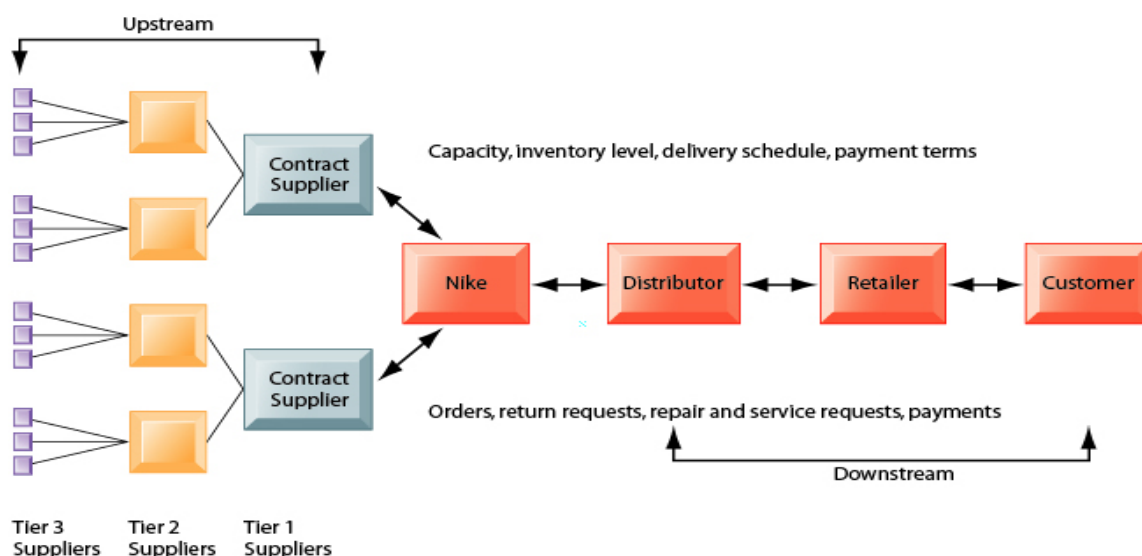
Enterprise systems produce value by increasing organizational efficiency and by providing firm wide information to help managers make better decisions. Enterprise systems create a foundation for a more customer-driven organization by integrating firm data to enable quicker responses to customer requests and information.

Enterprise systems:

- Use analytical tools to evaluate a firm's overall performance
- Use standard definitions, formats and performance figures across the organization

Supply Chain Management Systems

Supply chain management refers to the coordination of activities and involved in making and moving a product. The **supply chain** is the network of businesses and business processes involved the creation and selling of a product, from suppliers that procure raw materials through retail outlets and customers. The upstream portion of the supply chain includes the organization's suppliers and the processes for managing relationships with them. The downstream portion consists of the organizations and processes for distributing and delivering products to the final customers. The manufacturer also has internal supply chain processes for transforming the materials and services furnished by suppliers into finished goods.



NIKE'S

SUPPLY

CHAIN

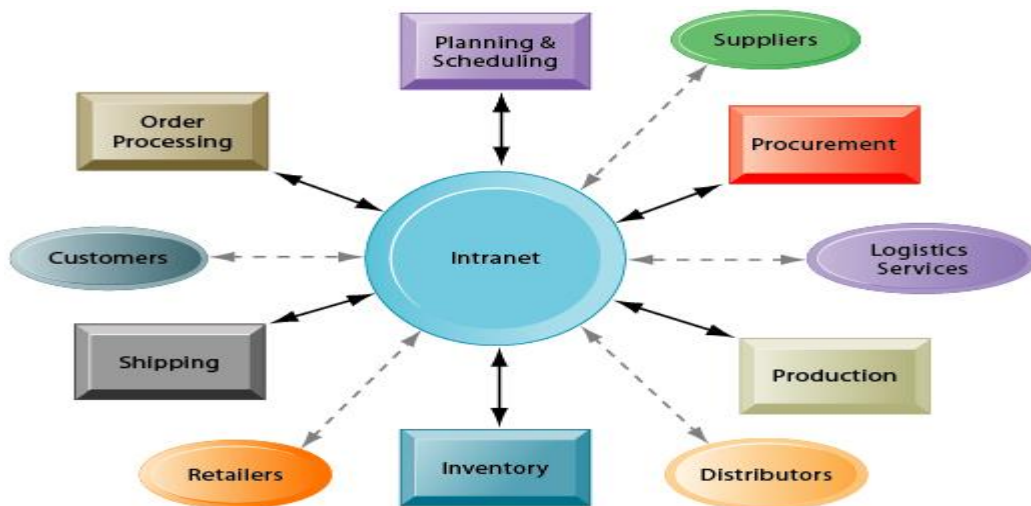
This figure illustrates the major entities in Nike's supply chain and the flow of information upstream and downstream to coordinate the activities involved in buying,

making and moving a product. Shown here is a simplified supply chain, with the upstream portion focusing only on the suppliers for sneakers and sneaker soles.

Supply chain software can be classified as either:

- **Supply chain planning systems:** Systems which enable the firm to generate demand forecasts for a product, develop sourcing and manufacturing plans for that product, make adjustments to production and distribution plans, and share that information with relevant supply chain members. One of the most important supply chain planning functions is **demand planning**, which determines how much product a business needs to make to satisfy all of its customers' demands.
- **Supply chain execution systems:** Systems that manage the physical flow of products through distribution centers and warehouses to ensure that products are delivered to the right locations in the most efficient manner.

Before the Internet, supply chain coordination was hampered by the difficulties of making information flow smoothly among disparate internal supply chain systems. Today, using intranets and extranets, all members of the supply chain can instantly communicate with each other, using up-to-date information to adjust purchasing, logistics, manufacturing, packaging, and schedules.



INTRANETS AND EXTRANETS FOR SUPPLY CHAIN MANAGEMENT

Intranets integrate information from isolated business processes within the firm to help manage its internal supply chain. Access to these private intranets can also be

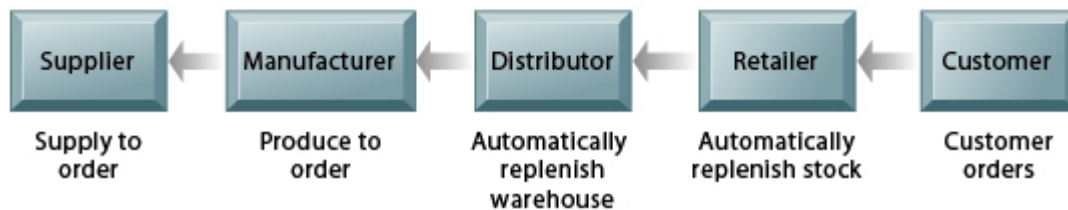
extended to authorized suppliers, distributors, logistics services, and, sometimes, to retail customers to improve coordination of external supply chain processes.

Earlier supply chain management systems were driven by a **push-based model** (also known as build-to-stock) in which production master schedules are based on forecasts or best guesses of demand for products, and products are "pushed" to customers. With Web-based tools, supply chain management follows a pull-based model (or demand-driven model or build-to-order), in which actual customer orders.

Push-Based Model



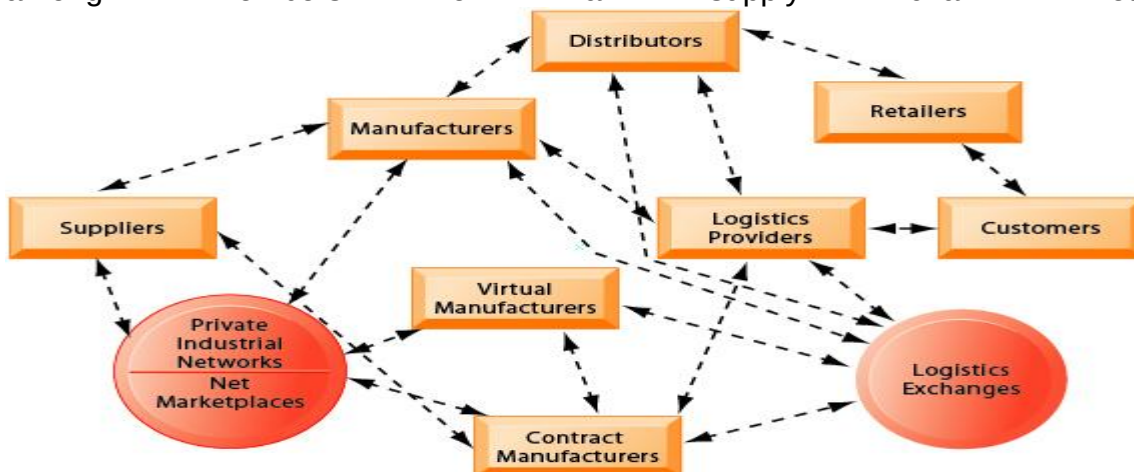
Pull-Based Model



PUSH-VERSUS PULL-BASED SUPPLY CHAIN MODELS

The difference between push- and pull-based models is summarized by the slogan "Make what we sell, not sell what we make."

Internet technology also makes it possible to move from sequential supply chains, where information and materials flow sequentially from company to company, to concurrent supply chains, where information flows in many directions simultaneously among members of a supply chain network.



THE FUTURE INTERNET-DRIVEN SUPPLY CHAIN

The future Internet-driven supply chain operates like a digital logistics nervous system. It provides multidirectional communication among firms, networks of firms, and e-marketplaces so that entire networks of supply chain partners can immediately adjust inventories, orders, and capacities.

The business value of supply chain management systems includes:

- Streamlined supply chain and accurate information
- Reduced supply chain costs
- Increased sales through accurate product availability

Customer Relationship Management Systems

Customers can be seen as an enterprise's most valuable asset, and customer relationship management systems enable large firms to understand and work with their customers.

CRM systems capture and integrate customer data from all over the organization, consolidating the data, analyzing the data, and then distributing the results to various systems and customer touch points across the enterprise. A **touch point** (also known as a contact point) is a method of interaction with the customer, such as telephone, e-mail, customer service desk, conventional mail.

Well-designed CRM systems provide a single enterprise view of the customer and provide customers with a single view of the company regardless of the touch point the customer uses.



CUSTOMER RELATIONSHIP MANAGEMENT (CRM)

CRM systems examine customers from a multifaceted perspective. These systems

use a set of integrated applications to address all aspects of the customer relationship, including customer service, sales and marketing.

Good CRM systems provide data and analytical tools for determining the financial lifetime value of a customer and customer loyalty and for identifying profitable customers.

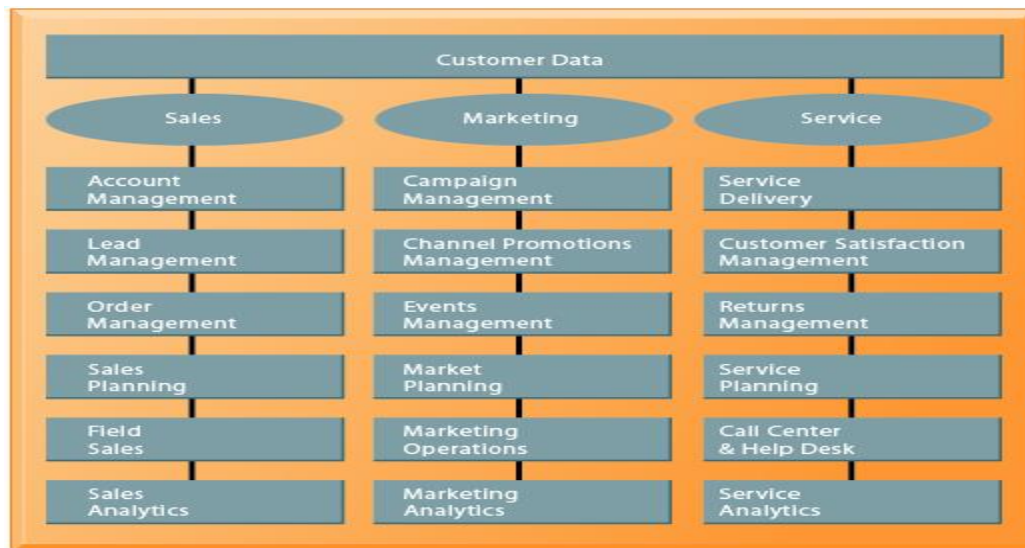
Commercial customer relationship management (CRM) software packages range from niche tools that perform limited functions, such as personalizing Web sites for specific customers, to large-scale enterprise applications. The more comprehensive CRM packages contain modules for:

- **Partner relationship management (PRM):** PRM software uses many of the same data, tools, and systems as customer relationship management to enhance collaboration between a company and its selling partners. It provides a company and its selling partners with the ability to trade information and distribute leads and data about customers, integrating lead generation, pricing, promotions, order configurations, and availability.
- **Employee relationship management (ERM):** ERM software deals with employee issues that are closely related to CRM, such as setting objectives, employee performance management, performance-based compensation, and employee training.

CRM systems typically provide software or tools for:

- **Sales force automation (SFA):** SFA modules help sales staff increase their productivity by focusing sales efforts on the most profitable customers. They provide sales prospect and contact information, product information, product configuration capabilities, and sales quote generation capabilities.
- **Customer service:** Customer service modules provide information and tools to make call centers, help desks, and customer support staff more efficient. They have capabilities for assigning and managing customer service requests and may include Web-based self-service capabilities.
- **Marketing:** Marketing modules support direct-marketing campaigns with capabilities for capturing prospect and customer data, qualifying leads, and scheduling and tracking campaign mailings. They include tools for analyzing marketing and customer data-identifying profitable and unprofitable customers, designing products and services to satisfy specific customer needs and interests, and identifying opportunities for cross-selling, up-selling, and bundling. **Cross-selling** is the marketing of complementary products to customers. Up-selling is the marketing of higher-value products or services to new or existing customers.

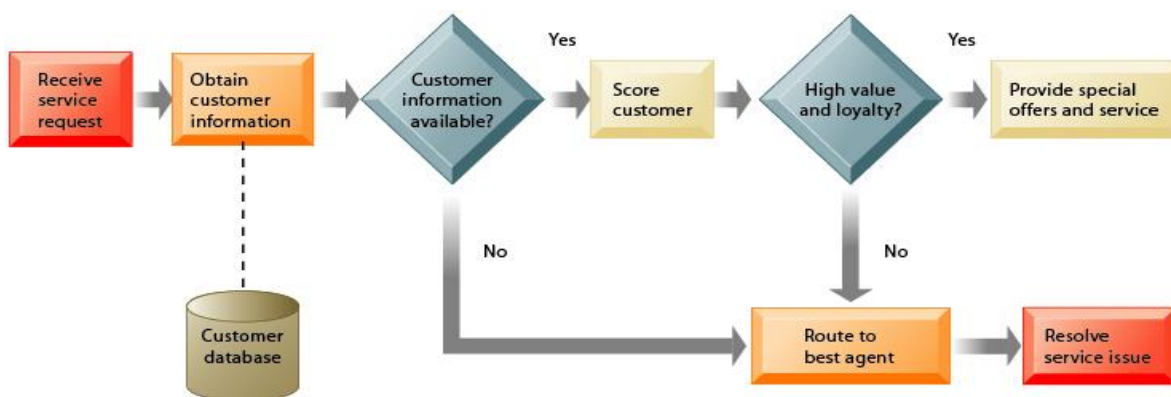
Bundling is cross-selling in which a combination of products is sold as a bundle at a price lower than the total cost of the individual products.



CRM SOFTWARE CAPABILITIES

The major CRM software products support business processes in sales, service, and marketing, integrating customer information from many different sources. Included are support for both the operational and analytical aspects of CRM.

CRM software can also be used to increase customer loyalty through customer service by identifying valued customers and providing them with special services or offers.

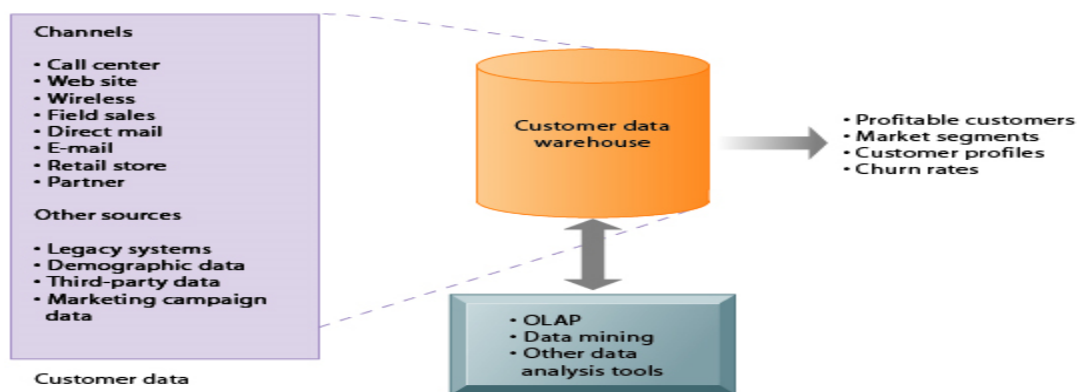


CUSTOMER LOYALTY MANAGEMENT PROCESS MAP

This process map shows how a best practice for promoting customer loyalty through customer service would be modeled by customer relationship management software. The CRM software helps firms identify high-value customers for preferential treatment.

CRM applications may support either:

- **Operational CRM:** Customer facing applications such as tools for sales force automation, call center and customer service support, and marketing automation.
- **Analytical CRM:** Applications that analyze customer data generated by operational CRM applications to provide information for improving business performance management. Analytical CRM applications are based on data warehouses that consolidate the data from operational CRM systems and customer touch points for use with online analytical processing (OLAP), data mining, and other data analysis techniques. An important output of analytical CRM is the **customer lifetime value (CLTV)**. CLTV is based on the relationship between the revenue produced by a specific customer, the expenses incurred in acquiring and servicing that customer, and the expected life of the relationship between the customer and the company.



ANALYTICAL CRM DATA WAREHOUSE
Analytical CRM uses a customer data warehouse and tools to analyze customer data collected from the firm's customer touch points and from other sources.

The business value of CRM systems includes

- Increased customer satisfaction
- Reduced direct-marketing costs
- More effective marketing
- Lower costs for customer acquisition and retention
- Increased sales revenues through identifying profitable customers

- **Reduced churn rate:** The **churn rate** measures the number of customers who stop using or purchasing products or services from a company and is an important indicator of the growth or decline of a firm's customer base.

Management Opportunities, Challenges and Solutions

Many firms obtain extraordinary business value from enterprise applications because of their power to improve process coordination and management decision making.

However, enterprise systems, supply chain management, and customer relationship management systems are very expensive to purchase and implement. Costs run even higher for organizations with global operations, which must manage organizational and technology changes in many different languages, time zones, currencies, and regulatory environments.

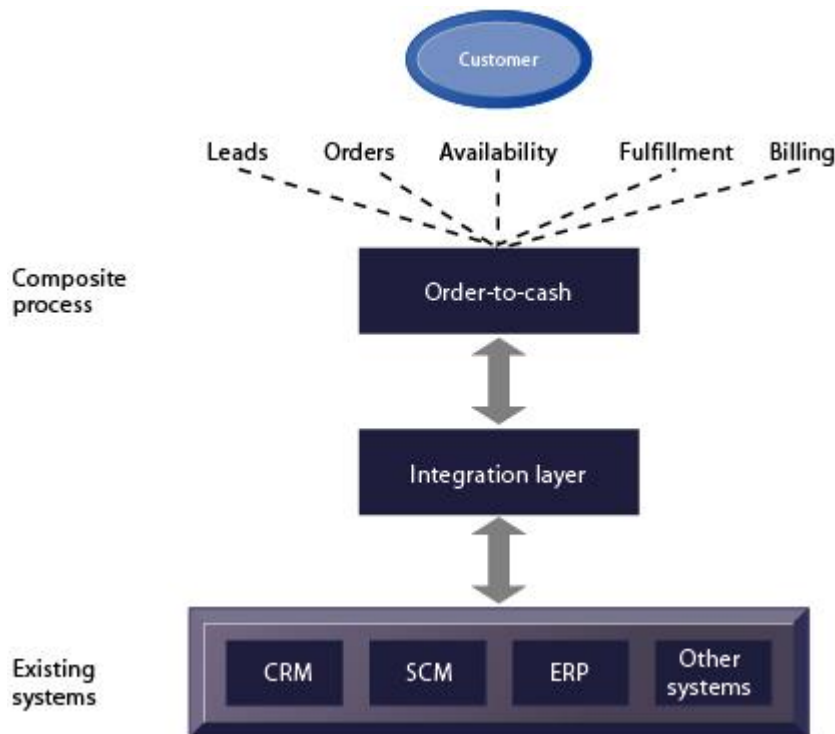
Enterprise applications require not only deep-seated technological changes but also fundamental changes in the way the business operates, including changes to business processes, employee responsibilities, and functions.

Enterprise applications also introduce "switching costs." Once an enterprise application is purchased and implemented, it becomes very costly to switch vendors.

Enterprise applications require defining and implementing standardized definitions of data throughout the organization.

Solutions for gaining more value from enterprise applications include:

- **Enterprise solutions (enterprise suites or e-business suites):** Flexible enterprise software that enables close linking between CRM, SCM, and enterprise systems, as well as to customer and supplier systems.
- **Service platforms:** A **service platform** integrates multiple applications from multiple business functions, business units, or business partners to deliver a seamless experience for the customer, employee, manager, or business partner. Enterprise-wide service platforms provide a greater degree of cross-functional integration than the traditional enterprise applications. To accomplish this, software tools (such as Web services and XML) use existing applications as building blocks for new cross-enterprise processes. Portal software can integrate information from enterprise applications and disparate in-house legacy systems, presenting it to users through a Web interface so that the information appears to be coming from a single source.



ORDER-TO-CASH

SERVICE

Order-to-cash is a composite process that integrates data from individual enterprise applications and legacy financial applications. The process must be modeled and translated into a software system using application integration tools.

Roles of Knowledge Management systems in Business

Knowledge management is the conscious process of defining, structuring, retaining, and sharing the knowledge and experience of employees within an organization.

As organizations evolve, expand into new areas, and define their approach to business, they develop significant [institutional knowledge](#). This information is invaluable to the company. Imparting it to new or less experienced staff is vital for maintaining successful operations.

The primary goal of knowledge management is facilitating the connection of staff looking for information, or institutional knowledge, with the people who have it.

With practical knowledge management in place, organizations can spread information and raise the level of expertise held by specific individuals or teams to improve the efficiency of their practices.

It often refers to training and learning in an organization or of its customers. It consists of a cycle of creating, sharing, structuring, and auditing knowledge to maximize the effectiveness of an organization's collective expertise.

Knowledge management in practice

Knowledge management can be separated into three main areas:

- Accumulating knowledge
- Storing knowledge
- Sharing knowledge

By accumulating and storing the staff's knowledge, companies hold onto what has made them successful in the past. In addition, sharing this information throughout the organization informs staff of past approaches that improve performance or better inform new strategies.

To achieve the goal of knowledge management, companies have to enable and promote a culture of learning and development, creating an environment where employees are encouraged to share information to better the collective workforce.

Benefits of knowledge management

A [survey](#) of over 286 people working in knowledge management across a range of industries, locations, and company sizes found the most significant benefits to be:

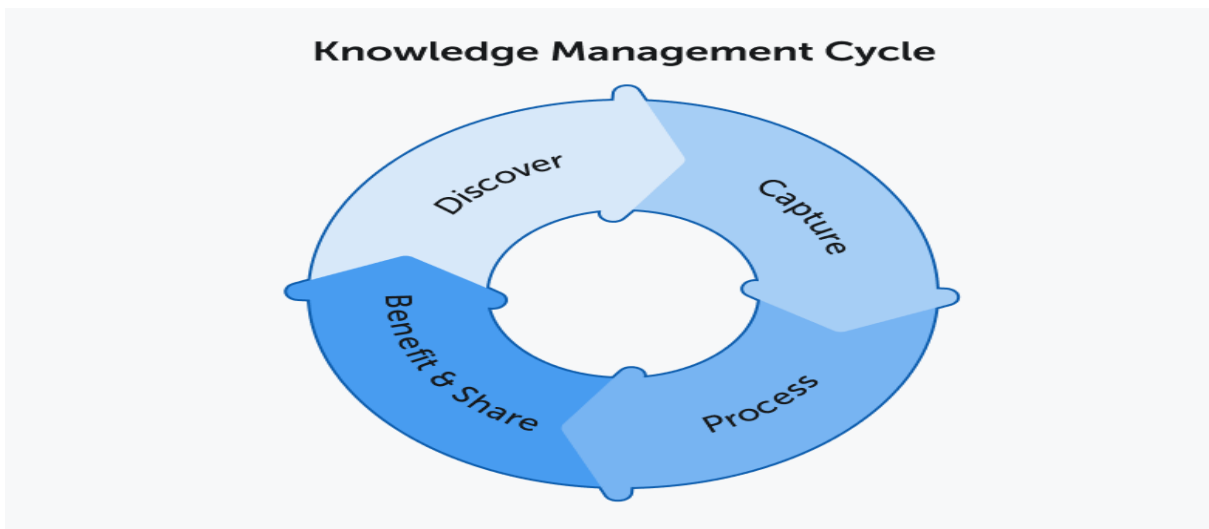
- Reduced time to find information
- Reduced time for new staff to become competent

- Reduced operational costs
- Improved customer satisfaction
- Improved bid win/loss ratio

Knowledge management process

Implementing effective knowledge management requires proactive strategies and incorporating multiple new processes.

Companies have to uncover the existing knowledge available to them, understand how to spread this information to produce additional value, and plan what this looks like in action.



Knowledge management process.

1. Discovery

Every organization has multiple sources of knowledge, from employees to data and records.

This could be the education and skillsets staff bring to the job, the experience and unique expertise they develop on the job, or hard drives of data that can positively affect the business with proper analysis.

During the discovery process, organizations must identify all the available sources of knowledge, with a particular emphasis on information that could be easily lost.

This process is simplified by a strong understanding of where and how knowledge flows around the organization.

2. Collection

Collecting all the available knowledge and data creates the foundation from which future processes build.

Sloppy or incorrect knowledge collection leads to decisions without a complete understanding of the organization and its capabilities.

Companies must audit their existing staff expertise, documentation, and external knowledge sources. A range of tools is available to help, including automated surveys, document scanning, and metadata.

Post-implementation, many organizations redefine internal processes to make capturing institutional knowledge a part of everyday processes. This could be through continual employee feedback systems or more in-depth off boarding procedures.

3. Assessment

This process involves the deep analysis of the knowledge gathered in the previous two steps. Data must be assessed and organized into a structured, searchable, and easily accessible form.

Assessment of the gathered knowledge is required to ensure it is accurate, offers value, and is up to date.

Then teams can determine how best to share information to improve company performance and give staff the knowledge they need to maximize performance.

Utilizing the right knowledge management system simplifies this process by allowing leadership to organize, assess, segment, and store a comprehensive knowledge database.

4. Sharing

The whole point of knowledge management is to give staff the expertise and information they need to do their job to the best of their ability.

Once you have built a detailed and accurate body of knowledge related to your company, you need to plan how it will be shared.

5. Application

This is the step where organizations reap the rewards of knowledge management. Discovering and storing institutional knowledge is just the beginning.

Staff utilizing newly acquired expertise in their tasks brings a range of benefits in productivity, accuracy, decision-making, and more innovative employees.

6. Creation

The final stage of knowledge management is to create more knowledge.

It should never be considered a one-and-done process. A single audit and rollout won't deliver the results you are looking for.

Knowledge management is a continual process that maximizes a company's performance for the expertise available to it.

What is artificial intelligence?

Artificial intelligence is a broad field, which refers to the use of technologies to build machines and computers that have the ability to mimic cognitive functions associated with human intelligence, such as being able to see, understand, and respond to spoken or written language, analyze data, make recommendations, and more.

Although [artificial intelligence](#) is often thought of as a system in itself, it is a set of technologies implemented *in* a system to enable it to reason, learn, and act to solve a complex problem.

What is machine learning?

[Machine learning](#) is a subset of artificial intelligence that automatically enables a machine or system to learn and improve from experience. Instead of explicit programming, machine learning uses algorithms to analyze large amounts of data, learn from the insights, and then make informed decisions.

Machine learning algorithms improve performance over time as they are trained—exposed to more data. Machine learning models are the output, or what the program learns from running an algorithm on training data. The more data used, the better the model will get.

Differences between AI and ML

Artificial intelligence

- AI allows a machine to simulate human intelligence to solve problems
- The goal is to develop an intelligent system that can perform complex tasks
- We build systems that can solve complex tasks like a human
- AI has a wide scope of applications

- AI uses technologies in a system so that it mimics human decision-making
- AI works with all types of data: structured, semi-structured, and unstructured
- AI systems use logic and decision trees to learn, reason, and self-correct

Machine learning

- ML allows a machine to learn autonomously from past data
- The goal is to build machines that can learn from data to increase the accuracy of the output
- We train machines with data to perform specific tasks and deliver accurate results
- Machine learning has a limited scope of applications
- ML uses self-learning algorithms to produce predictive models
- ML can only use structured and semi-structured data
- ML systems rely on statistical models to learn and can self-correct when provided with new data

Applications of AI in business management include:

- spam filters.
- smart email categorisation.
- voice to text features.
- smart personal assistants, such as Siri, Cortana and Google Now.
- automated responders and online customer support.
- process automation.
- sales and business forecasting.
- security surveillance.

Knowledge work system

It is a specific management system created for engineers, scientists, and other intelligent people. Such efficient knowledge management applications provide an online directory where users may look for company employees who can provide them with the precise information they require.

Requirements of Knowledge Work Systems

Knowledge work systems have characteristics that reflect the special needs of knowledge workers. First, knowledge work systems must give knowledge workers the specialized tools they need, such as powerful graphics, analytical tools, and communications and document management tools. These systems require great computing power to handle the sophisticated graphics or complex calculations necessary for such knowledge workers as scientific researchers, product designers, and financial analysts. Because knowledge workers are so focused on knowledge in the external world, these systems also must give the worker quick and easy access to external databases.

A user-friendly interface is very important to a knowledge worker's system. User friendly interfaces save time by enabling the user to perform needed tasks and get to required information without having to spend a lot of time learning how to use the computer. Saving time is more important for knowledge workers than for most other employees because knowledge workers are highly paid—wasting a knowledge worker's time is simply too expensive, and knowledge workers can easily fall prey to information overload (Farhoomand and Drury, 2002). Figure 12-10 summarizes the requirements of knowledge work systems.

